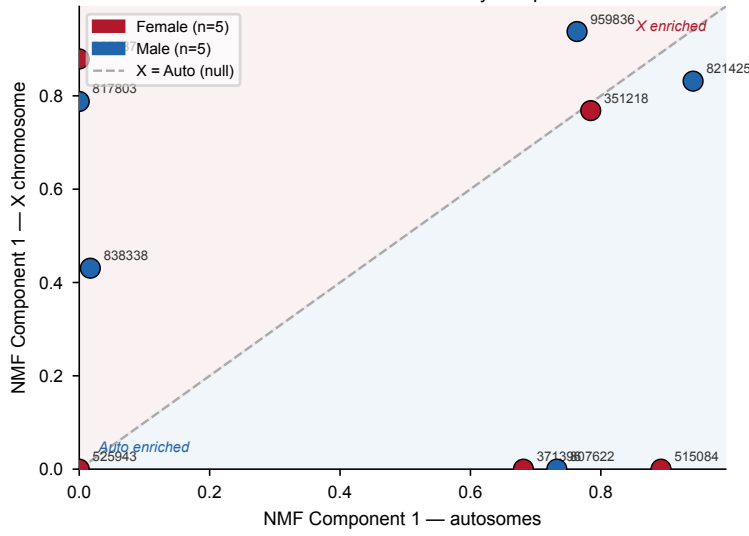
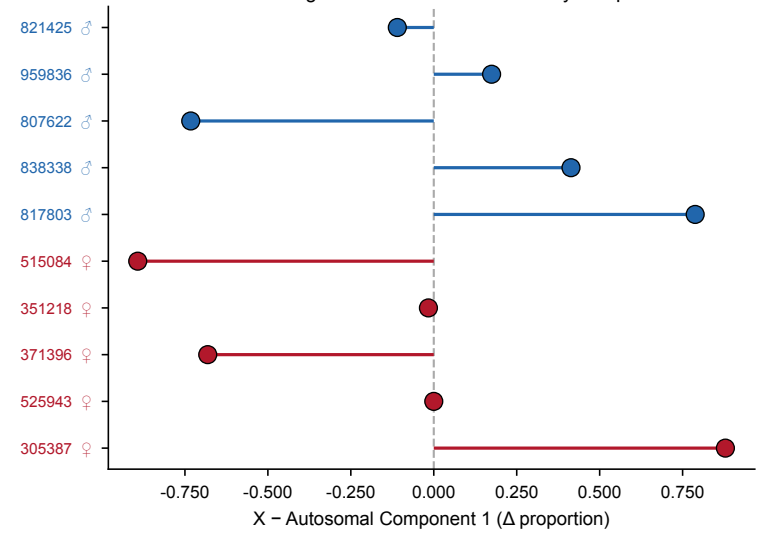


a

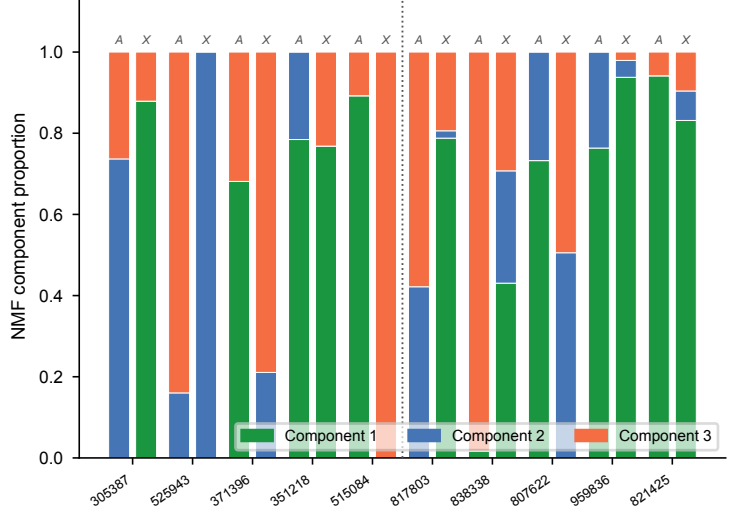
Autosomal vs X-linked ancestry component

**b**

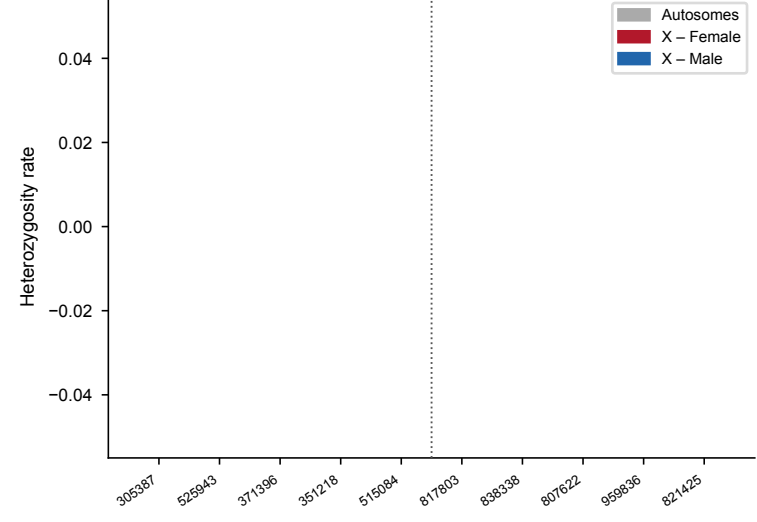
Sex-biased signal: X vs autosomal ancestry component

**c**

NMF ancestry decomposition: autosomes (A) vs X (X)

**d**

Heterozygosity: autosomes vs X chromosome (real males: X het < auto het; mock data: no difference expected)



Note (mock data, -alt-frequency 0.5): X het = auto het = 0 for all samples